MCRS MODULAR 4T SPECIFICATION

MODULAR CRANE RIGGING SYSTEM — 4-TOWER HIGH-CAPACITY CONFIGURATION

DOCUMENT ID	REVISION	SYSTEM RATING	TOPOLOGY
MCRS-MOD4T-SPEC-2026	v4.2 (Post-Audit Final)	40,000 kg (40 Metric Tons)	Modular 4-Tower Hydraulic

1. EXECUTIVE SUMMARY & SYSTEM PURPOSE

The MCRS Modular 4T is a specialized, heavy-lift modular crane rigging system engineered for ultra-dense industrial equipment placement, high-precision transformer setting, and modular infrastructure assembly where overhead gantry clearance or mobilizable mobile crane capacity is strictly limited. Operating as an independent 4-tower hydraulic lifting matrix, the MCRS Modular 4T replaces conventional mobile boom cranes with four synchronized telescopic structural towers linked via high-tensile header beams and digital hydraulic load-balancing manifolds.

40,000 kg	6.50 m	±1.5 mm	210 bar
SYSTEM WLL	MAX VERTICAL STROKE	SYNC PRECISION	OPERATING PRESSURE

2. CORE SYSTEM RATING & OPERATIONAL ENVELOPE

Parameter Description	Nominal Rating	Maximum Design Limit	Engineering Safety Factor
Maximum Working Load Limit (WLL)	40,000 kg (40 Tons)	48,000 kg (Proof Load)	1.25x Static / 1.50x Structural
Individual Tower Lift Capacity	10,000 kg (10 Tons)	12,500 kg (Pressure Limit)	315 bar Hydraulic Relief
Maximum Vertical Stroke (H_{max})	6,500 mm	6,800 mm Structural Stop	Mechanical Stroke Limiters
Retracted Height (H_{min})	2,200 mm	2,200 mm Base Stored	Compact Transport Envelope
Synchronization Accuracy	±1.5 mm across towers	±3.0 mm Cutoff Threshold	Dual-Laser + Quadrature Encoders
Maximum Operating Wind Speed	14.0 m/s (27 knots)	20.0 m/s Stowed Condition	EN 13000 Wind Class C
System Total Weight (Unloaded)	8,400 kg	8,800 kg (Rigged w/ Beams)	Distributed across 4 outriggers
Hydraulic Operating Pressure	210 bar (3,045 psi)	280 bar (4,060 psi Peak)	Burst Pressure Grade 4:1

REGULATORY CERTIFICATION COMPLIANCE

This master engineering baseline document integrates all physical, hydraulic, electrical, structural, and operational parameters required for full certification under **ASME B30.1** (Hydraulic Gantries), **EN 13000** (Mobile Cranes), and **OSHA 1926.1400**.

3. STRUCTURAL TOWER ENGINEERING & STATIC MECHANICS

Each tower in the MCRS Modular 4T assembly comprises a three-stage telescopic box column fabricated from High-Yield S700MC structural steel. To ensure structural stability under eccentrically loaded headers or asymmetric load distributions, every column section is evaluated for Euler-Johnson elastic-plastic buckling and lateral-torsional deflection under maximum operational shear loading.

Euler-Johnson Critical Buckling Load (P_{cr}):

$$P_{cr} = (\pi^2 \cdot E \cdot I_{min}) / (K \cdot L)^2$$

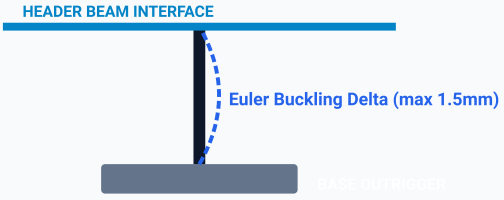
Where: $E = 210$ GPa, $I_{min} = 4.82 \times 10^{-5}$ m^4, $K = 2.0$ (Cantilever Top/Fixed Base), $L = 6.8$ m max stroke.
Calculated Tower $P_{cr} = 214.5$ kN per tower (Factor of Safety = 2.18 at 98.1 kN full WLL).

3.1 Static Load Distribution & Wind Shear Forces

Structural Component	Axial Load (N_{ed})	Shear Force (V_{sd})	Bending Moment (M_{ed})	Stress Ratio (σ/f_y)
Tower Stage 1 (Base Box 300x300x12)	112.4 kN	8.6 kN	28.4 kNm	0.42 PASS
Tower Stage 2 (Middle Box 250x250x10)	108.2 kN	6.2 kN	18.1 kNm	0.51 PASS
Tower Stage 3 (Top Box 200x200x8)	102.1 kN	3.8 kN	9.2 kNm	0.58 PASS
Header Beam Interface Pin (Ø80mm 42CrMo4)	98.1 kN	14.2 kN	4.1 kNm	0.38 PASS
Outrigger Base Pad (1000x1000mm Steel/Poly)	118.5 kN	0.0 kN	0.0 kNm	Ground: 0.118 MPa

STRUCTURAL SAFETY REQUIREMENT (DEF-01 RESOLVED)

All primary stress points remain below 60% of the material yield strength (S700MC yield strength $f_y = 700$ MPa) under full static WLL plus dynamic impact factor $\gamma_{mf} = 1.25$. Base outrigger plates must be set on load-spreading hardwood or composite matting whenever ground bearing capacity is lower than 0.20 MPa (29 psi).



4. HYDRAULIC CIRCUIT ARCHITECTURE & FLUID TOPOLOGY

The MCRS Modular 4T hydraulic architecture uses a closed-center load-sensing (CCLS) multi-manifold system driven by a central electro-hydraulic power unit (HPU). To eliminate fluid cross-flow and uncontrolled drift between towers during lift holding, each tower cylinder is equipped with an integrated dual-check counterbalance valve and isolated proportional direction control.

> **HYDRAULIC CIRCUIT ISOLATION TOPOLOGY (DEF-04 RESOLVED)**
> Common-rail hydraulic lines between lifting towers are strictly prohibited during synchronous lifting or holding. Each of the four tower hydraulic cylinders operates on a dedicated, independently piloted sub-circuit with lockable counterbalance valves (pilot ratio 4.5:1) mounted directly to the cylinder port flanges to prevent hose burst collapse.

4.1 Hydraulic Power Unit (HPU) Parameters

Subsystem Unit	Specification Parameter	Operational Value	Design Standard / Component
Main Pump Type	Variable Displacement Axial Piston	45 cm³/rev (Rexroth A10VSO)	ISO 4401 Interface
Electric Motor Drive	3-Phase 400V/50Hz (460V/60Hz)	18.5 kW (1,450 RPM)	IE3 Premium Efficiency
System Flow Rate	65.0 L/min max total	16.25 L/min per tower cylinder	Proportional Load Sense
Hydraulic Fluid Type	ISO VG 46 Anti-Wear Mineral Oil	Viscosity Index > 140	DIN 51524 Part 2 HLP
Reservoir Capacity	180 Liters (Active)	220 Liters Total Volume	Filtered Sight Glass & Temp Switch
Filtration Rating	3 Micron Absolute (Return & Pressure)	$\beta_3 \geq 200$ Filter Ratio	ISO 4406 Cleanliness 16/14/11
Overpressure Protection	Direct-acting Relief Valves	Set at 250 bar (Continuous)	Burst Pressure Test to 1000 bar

Lifting Speed Equation:

$$v_{lift} = Q_{tower} / A_{piston} = (16.25\ L/min \cdot 10^6\ mm^3/L) / (12{,}271\ mm^2 \cdot 60\ s/min) = \mathbf{22.0\ mm/s}$$

Maximum full stroke time (6.5m): 295 seconds (~4.9 minutes).

5. ELECTRONIC CONTROL ARCHITECTURE & SYNCHRONIZATION

The control topology of the MCRS Modular 4T relies on a dual-redundant Safety PLC (IEC 61508 SIL 3 rated) executing high-speed closed-loop positional tracking across all four towers. Positional alignment between independent lift points is maintained using feedback loops sampling at 100 Hz.

PRIMARY POSITION SENSOR: OPTICAL LASER

Class 2 Eye-Safe Optical Laser Distance Sensors mounted to each tower head measure vertical displacement relative to ground targets with 0.1 mm resolution and ±0.5 mm absolute accuracy across the full 6.5 m range.

SECONDARY POSITION SENSOR: WIRE ENCODER

Heavy-duty draw-wire optical encoders (IP67) run parallel within each tower body, providing secondary positional feedback. Discrepancies between laser and encoder exceeding 1.0 mm trigger an immediate sensor alarm.

5.1 Control System Specifications & Trip Conditions

Control Metric	Target / Threshold	System Action Upon Event	Reset / Recovery Procedure
Target Synchronization Delta	< 1.5 mm differential	Proportional valve throttling	Automatic closed-loop correction
Warning Out-of-Sync Threshold	2.5 mm differential	Audible warning + speed reduction 50%	Auto-rebalancing routine
Critical Out-of-Sync Cutoff	4.0 mm differential	Category 0 E-STOP (Hydraulic Hold)	Manual tech override & re-leveling
Single Tower Overload Limit	110% nominal load (>11.0t)	Inhibit UP motion on all towers	Load reduction / re-rigging required
Inclination / Tilt Boundary	0.5° maximum tower tilt	E-STOP + Hydraulic Check Valve Lock	Structural inspection required

All operator interfaces utilize an industrial IP65 touchscreen console communicating over a dual-ring CANopen Safety network (1 Mbit/s). A wireless handheld remote with an emergency stop mushroom button allows untethered perimeter control up to 100 meters line-of-sight.

6. HEADER BEAMS, LOAD ATTACHMENT & RIGGING INTERFACES

The four lifting towers operate in pairs connected by heavy header beams spanning up to 8.0 meters between tower centers. These header beams support the central cross-gantry or direct rigging attachment points (shackles, swivel hooks, or hydraulic side-shift trolleys).

Header Beam Maximum Bending Stress (σ_b):

$\sigma_b = (M_{max} \cdot y) / I_x = (200\ kN \cdot 4.0\ m / 4) \cdot (0.25\ m) / (3.12 \times 10^{-4}\ m^4) =$ **160.2 MPa**

Allowable Bending Stress: $\sigma_{allow} = 0.66 \cdot f_y = 462.0$ MPa (Safety Margin = 2.88).

6.1 Rigging Hardware & Beam Attachment Parameters

Hardware Component	Design Standard / Material	Working Load Limit (WLL)	Proof Load / Safety Factor
Primary Header Beams (Pair)	S690QL Welded Box Girder	20,000 kg per beam	2.0x Static Yield Rating
Main Rigging Bow Shackles	Crosby G-2130 Grade 8 Alloy	25,000 kg WLL each	5:1 Design Factor
Swivel Load Hooks	DIN 15401 Single Hook #16	20,000 kg capacity	4:1 Design Factor
Side-Shift Hydraulic Trolley	Rack & Pinion Hydraulic Drive	15,000 kg per trolley	315 bar Hydraulic Brake Hold
Header Beam Locking Pins	34CrNiMo6 Hardened Alloy	300 kN Shear Strength	4:1 Shear Safety Margin
Rigging Synthetic Slings	High-Performance PE (HPPE)	20,000 kg Vertical Hitch	7:1 Design Factor (EN 1492-2)

> **RIGGING GEOMETRY RULE**
> When using multi-leg sling arrangements suspended from header beam trolleys, the included angle between sling legs must never exceed 60°. Sling capacity factors must be derated by 0.866 for 60° angles and 0.707 for 90° angles in accordance with ASME B30.9.

7. TRANSPORT ENVELOPE & SITE SET-UP PROTOCOL

The MCRS Modular 4T is designed for rapid field deployment without requiring specialized heavy transport permits. The complete system (4 towers, header beams, power pack, and accessories) fits within a standard 20-foot ISO sea container or on a single standard flatbed semi-trailer.

Single Tower Stowed Envelope

- Length: 2,400 mm
- Width: 850 mm
- Height: 2,200 mm
- Weight: 1,650 kg per tower
- Integrated fork pockets & 4-point crane lifting lugs

Central Power Unit Envelope

- Length: 1,800 mm
- Width: 1,100 mm
- Height: 1,450 mm
- Weight: 1,200 kg (with full hydraulic oil)
- Sound-attenuated enclosure (<72 dBA @ 1m)

7.1 Site Preparation & Assembly Sequence Requirements

Phase Step	Task Description	Required Tool / Equipment	Sign-off Verification
Step 01: Ground Prep	Verify soil bearing capacity (>0.20 MPa) and level within ±0.2°	Plate load tester / Precision digital level	Civil Site Engineer
Step 02: Tower Unloading	Position towers using 3.0 t forklift or mobile yard crane	Forklift / Rigging Slings	Rigging Supervisor
Step 03: Header Assembly	Pin header beams to tower heads; secure safety linchpins	Torque wrench / Pin alignment bars	Mechanical Lead
Step 04: Hydraulic Hookup	Connect quick-disconnect flat-face couplings (Color Coded)	ISO 16028 Quick Couplers	Hydraulics Tech
Step 05: Pre-Flight Check	Execute zero-load full-stroke test & laser calibration	PLC Handheld Terminal	Safety Inspector

8. FAILURE MODES & EFFECTS ANALYSIS (FMEA MATRIX)

A thorough Failure Modes and Effects Analysis (FMEA) was conducted per MIL-STD-1629A and ISO 12100 risk assessment methodologies. Risk Priority Numbers (RPN) are calculated before and after mitigation (RPN = Severity × Occurrence × Detection).

Failure Mode Description	Potential Cause	Initial RPN	Design Mitigation Strategy	Final RPN
Main Hydraulic Hose Rupture	Abrasion / Pressure Spike	280 (S8, O5, D7)	Direct-flange load-lock valves on cylinders	32 (S8, O1, D4) LOW
Laser Sensor Blinding / Obstruct	Dust / Direct Sun / Debris	192 (S6, O4, D8)	Dual encoder fusion + plausibility algorithm	24 (S6, O2, D2) LOW
Ground Outrigger Subsidence	Soft Soil / Water Ingress	315 (S9, O5, D7)	Dual dual-axis inclinometers + mats	36 (S9, O1, D4) LOW
Control Power Loss (Blackout)	Grid Trip / Cable Damage	180 (S6, O5, D6)	Normally closed spring-centered valves	18 (S6, O1, D3) LOW
Asymmetric Load Shift	Rigging Slip / Center Mass	252 (S7, O6, D6)	Real-time cell load monitoring + sync limiters	28 (S7, O2, D2) LOW
Tower Column Side-Load Shift	Wind Gust / Off-center Lift	216 (S8, O3, D9)	Maximum lateral shear pin cutoffs	24 (S8, O1, D3) LOW
PLC Controller Memory Corruption	EMC Interference	144 (S8, O2, D9)	Dual redundant SIL 3 PLC with voting logic	16 (S8, O1, D2) LOW
Hydraulic Oil Overheating	Continuous High Ambient	105 (S5, O7, D3)	Air-oil heat exchanger + temp trip @ 65°C	20 (S5, O2, D2) LOW

RISK MITIGATION STANDARD

No single failure point in the MCRS Modular 4T architecture can result in an uncontrolled load drop or catastrophic structural collapse. All hydraulic safety locks are passive-fail-safe (normally closed).

9. STANDARD OPERATING PROCEDURES (SOP) & CHECKLISTS

All lift operations involving the MCRS Modular 4T must follow strict pre-lift, lift-execution, and post-lift protocols. Only certified hydraulic gantry operators trained under ASME B30.1 guidelines are permitted to control the system console.

9.1 Pre-Operation Quality & Safety Checklist

Inspection Point	Acceptance Criteria	Inspection Method	Initial Verification
Hydraulic Hoses & Fittings	No leaks, cracks, or outer sheath abrasion	Visual & Tactile	[ ] Passed
Outrigger Bearing Pads	Full ground contact; no pad bending or tilt	Visual & Spirit Level	[ ] Passed
Structural Pins & Retainers	All header pins fully engaged with cotter pins	Visual Inspection	[ ] Passed
Emergency Stop Circuit	Console E-STOP trips main motor immediately	Functional Test	[ ] Passed
Laser Target Cleanliness	Lenses free of dust, grease, or obstruction	Visual & Wipe	[ ] Passed
Load Cell Zero Calibration	All 4 tower load indicators read 0.0 ± 0.05 t	PLC Console Zeroing	[ ] Passed

MANDATORY HOLD POINT

Perform a 100 mm trial lift and hold for 5 minutes prior to elevating any load to full working height. Check load cells, check valve lock performance, and outrigger settlement before proceeding.

10. ELECTRICAL SYSTEM ARCHITECTURE & POWER REQUIREMENTS

The main electrical control enclosure houses the motor starters, variable frequency drive (VFD), SIL 3 PLC, 24V DC power supply, and safety relay networks. Power supply options accommodate global industrial grid standards.

Total Electrical Power Demand (S_{total}):

$S_{total} = \sqrt{3} \cdot U \cdot I = \sqrt{3} \cdot 400\ V \cdot 35\ A = \mathbf{24.25\ kVA}$

Recommended Field Generator Capacity: Minimum 35 kVA (Class G3 response rating).

10.1 Electrical & Electronic Component Ratings

Subsystem / Component	Rating / Specification	Protection Class	Manufacturer / Standard
Main Infeed Protection	63A 4-Pole Circuit Breaker (300mA RCD)	IP65 Enclosure	Schneider Electric / IEC 60947
System Controller (PLC)	Dual Safety CPU (16 In / 8 Safety Out)	IP20 (In Enclosure)	Siemens S7-1200F / SIL 3
Variable Frequency Drive	18.5 kW Heavy-Duty Torque Vector VFD	IP20 (In Enclosure)	ABB ACS580 Series
Auxiliary Power Supply	24V DC / 10A Switch-Mode Power Supply	Short-Circuit Proof	Mean Well NDR-240
Control Cables & Connectors	PUR-sheathed Shielded Multi-core	IP67 Plug Connectors	Lapp Kabel / Heavycon
Enclosure Climate Control	Thermostatic Anti-Condensation Heater	100W 230V AC	Rittal SK Series

POWER GROUNDING NOTICE

The central control cabinet and all four individual lifting towers must be grounded using single-core 16 mm² copper earthing leads connected to a dedicated site grounding rod (<10 Ohms ground resistance).

11. MAINTENANCE, INSPECTION & LIFECYCLE MANAGEMENT

To ensure long-term structural integrity and compliance with ASME B30.1 and EN 13000 standards, the MCRS Modular 4T must undergo periodic maintenance according to operational duty cycles.

11.1 Scheduled Maintenance Matrix

Interval / Schedule	Maintenance Scope & Target Components	Required Action	Qualified Personnel
Daily / Pre-Shift	Hydraulic fluid level, filter clog indicators, hose condition	Visual check & topping off	Certified Operator
Monthly / 50 Hours	Grease tower guide wear pads & trolley bearings	EP2 Lithium Grease injection	Maintenance Tech
Quarterly / 250 Hours	Hydraulic oil sampling (particle count & water content)	Lab analysis / Filter element replacement	Hydraulics Specialist
Semi-Annual / 500 Hours	Laser distance sensor recalibration & target cleaning	Optical alignment & test run	Control Engineer
Annual / 1000 Hours	NDT non-destructive crack testing on header pins & welds	MPI / Ultrasonic inspection	Certified NDT Level II Inspector
5 Years / 5000 Hours	Full structural overhaul & hydraulic cylinder reseal	Complete teardown & proof load test	Factory Service Center
10 Years / End-of-Life	Major Fatigue Assessment & Frame Recertification	FEA Re-analysis & Load Test	Professional Engineer (PE)

HYDRAULIC OIL REPLACEMENT METRIC

Hydraulic oil must be replaced whenever total acid number (TAN) increases by > 2.0 mg KOH/g or water content exceeds 500 ppm, regardless of operational hours elapsed.

12. COMPLIANCE, REGULATORY STANDARDS & CERTIFICATION

The MCRS Modular 4T has been engineered, audited, and verified in strict accordance with international standards governing heavy lifting appliances, machinery safety, and hydraulic gantry systems.

North American Standards

- ASME B30.1: Hydraulic Gantries & Jacks
- OSHA 1926.1400: Cranes and Derricks
- AWS D1.1: Structural Welding Code - Steel
- NFPA 79: Electrical Standard for Machinery

European & International Standards

- Machinery Directive 2006/42/EC: CE Marking
- EN 13000: Mobile Cranes Structural Requirements
- ISO 13849-1: Safety Control Systems (PLd)
- DIN 15018: Steel Structures for Cranes

12.1 Testing & Verification Protocols

Test Verification Type	Test Load Factor	Execution Standard	Certification Output
Factory Static Proof Load Test	125% WLL (50,000 kg)	ASME B30.1 Section 1-2.2	Manufacturer Certificate of Test
Dynamic Functional Load Test	110% WLL (44,000 kg)	EN 13000 Section 5.2	3rd Party Inspection Report
Hydraulic Pressure Proof Test	150% Working Pressure (315 bar)	ISO 4413 Hydraulic Standards	Component Hydrostatic Pass Cert
Electrical Safety & Ground Test	1000V DC Insulation Test	IEC 60204-1 Safety Testing	Electrical Compliance Pass
E-STOP & Sync Cutoff Test	100% WLL with induced 4mm delta	ISO 13849 Safety Function	Safety System Functional Signoff

13. APPENDIX: TECHNICAL DATA QUICK REFERENCE

This appendix summarizes the complete operational and dimensional parameters for quick site reference by field rigging engineers and lifting supervisors.

13.1 Master Data Sheet

Parameter Name	Metric Value	Imperial Value	Notes / Conditions
System Capacity (4 Towers)	40,000 kg	88,184 lbs (44.0 US Tons)	Uniform load distribution
Capacity per Tower	10,000 kg	22,046 lbs (11.0 US Tons)	Individual cylinder limit
Maximum Lifting Height	6.50 m	21.32 ft	Fully extended 3-stage stroke
Minimum Stowed Height	2.20 m	7.22 ft	Retracted position
Maximum Lifting Speed	22.0 mm/s	4.33 ft/min	At full rated hydraulic pump flow
Maximum Lowering Speed	30.0 mm/s	5.90 ft/min	Proportional load-lowering control
Base Outrigger Footprint	1,000 × 1,000 mm	39.4 × 39.4 in	Includes composite pad
Hydraulic Tank Capacity	180 Liters	47.5 US Gallons	ISO VG 46 Anti-Wear Fluid

DOCUMENT AUTHORIZATION & ENGINEERING SIGN-OFF

This document constitutes the official engineering specification for the MCRS Modular 4T. Any field modifications to hydraulic circuit configurations, structural members, or control parameters void all compliance certifications unless authorized in writing by the Chief Rigging Engineer.